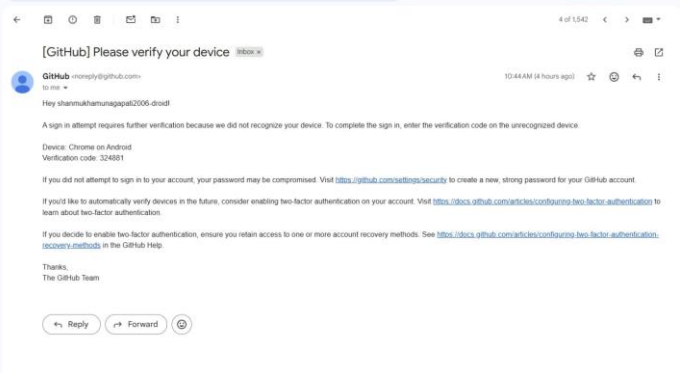


Ex.No.04 - Email Header Analysis and Spoofing Detection using MHA

Experiment Name	Email Header Analysis and Spoofing Detection using MHA
Aim	To analyse email headers using Mail Header Analyzer (MHA) and identify possible email spoofing by examining email routing information and authentication mechanisms such as SPF, DKIM, and DMARC.
Tool Used	Mail Header Analyzer (MHA), Gmail, Web Browser, Email Header

Experiment Focus: Obtaining the original header, analysing Received headers, checking SPF/DKIM/DMARC, examining Message-ID and Return-Path, and identifying possible spoofing anomalies.

Output 1 - GitHub email in Gmail



This screen shows the selected GitHub email in Gmail. The message provides the initial email content and sender context required before obtaining the complete original header.

Output 2 - Original Message and SPF/DKIM/DMARC

Original message	
Message ID	<6a8fc7b54a49a_1eee11b86548d@accountsecurityworker-97cdf9c7-gpq54.mail>
Created at:	Thu, Aug 27, 2026 at 10:44 AM (Delivered after 1 second)
From:	GitHub <noreply@github.com>
To:	shanmukhamunagapati2006-droid <shanmukhamunagapati2006@gmail.com>
Subject:	[GitHub] Please verify your device
SPF:	PASS with IP 192.30.252.205 Learn more
DKIM:	'PASS' with domain github.com Learn more
DMARC:	'PASS' Learn more
Download Original Copy to clipboard	

The Gmail Original Message view displays the Message ID, creation time, From, To, Subject, and authentication results. SPF, DKIM, and DMARC are shown for the message.

Forensic significance: The initial Gmail and Original Message outputs establish the source message and expose the authentication fields required for header analysis.

Output 3 - Mail Header Analyzer input

 **Email Header Analyzer**

Paste Header:

If you did not attempt to sign in to your account, your password may be compromised. Visit <https://github.com/settings/security> to create a new, strong password for your GitHub account.

If you'd like to automatically verify devices in the future, consider enabling two-factor authentication on your account. Visit <https://docs.github.com/articles/configuring-two-factor-authentication> to learn about two-factor authentication.

If you decide to enable two-factor authentication, ensure you retain access to one or more account recovery methods. See <https://docs.github.com/articles/configuring-two-factor-authentication-recovery-methods> in the GitHub Help.

Thanks,
The GitHub Team

[Analyze Header](#)

 Loading

ABOUT EMAIL HEADERS

This tool will make email headers human readable by parsing them according to RFC 822. Email headers are present on every email you receive via the Internet and can provide valuable diagnostic information like hop delays, anti-spam results and more. If you need help getting copies of your email headers, [just read this tutorial](#).

The copied email header was pasted into Mail Header Analyzer. The analyzer input stage prepares the complete header for structured parsing and forensic examination.

Output 4 - Header analysis / raw header

[illegible]

The raw-header analysis displays detailed message and routing information. Header fields are examined to understand the email path and authentication-related data.

Forensic significance: The MHA input and raw-header outputs provide the detailed routing and metadata needed to trace the message through mail infrastructure.

Output 5 - Received Header

Received Header

[illegible]

The Received header information was reviewed to examine hop number, submitting and receiving hosts, time, delay, connection information, IP address, and hostname details.

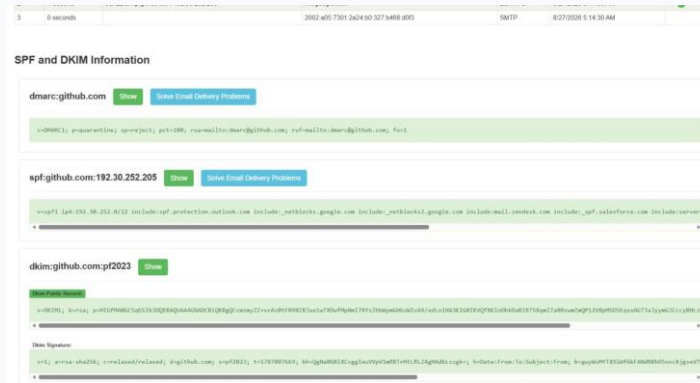
Output 6 - Headers Found

[illegible]

The Headers Found table lists important header fields and their values, including Delivered-To, Received, ARC information, Authentication-Results, DKIM-Signature, Message-ID, and Return-Path.

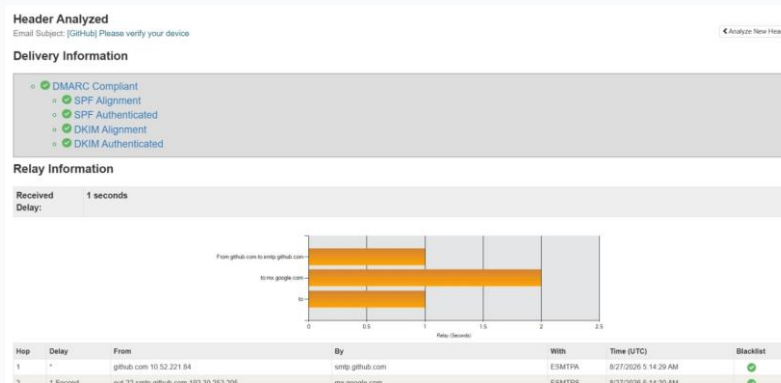
Forensic significance: The Received and Headers Found outputs allow comparison of routing, IP, Message-ID, Return-Path, and other header information.

Output 7 - SPF and DKIM Information



The authentication section provides SPF, DKIM, and DMARC information. The supplied analysis records successful authentication for the examined message.

Output 8 - Header Analyzed / Delivery Information



The final MHA view summarizes authentication and relay information. The delivery path and authentication indicators were reviewed together to assess possible spoofing anomalies.

Observation

The email header was successfully obtained from Gmail and analysed using Mail Header Analyzer. The MHA analysis displayed routing information, Received headers, Return-Path, Message-ID, SPF information, DKIM signatures, and authentication results. The supplied results record SPF: PASS, DKIM: PASS, and DMARC: PASS.

Result

The email header was successfully analysed using MHA. SPF, DKIM, and DMARC were PASS, and no obvious evidence of email spoofing was identified from the analysed header.

Conclusion

The experiment demonstrated how email headers can be analysed for authenticity and routing. SPF, DKIM, and DMARC are important authentication mechanisms, while Received headers, IP addresses, Message-ID, and Return-Path provide useful information for email forensic investigation.

Key forensic keywords

[MHA](#) | [Email Header](#) | [Received Header](#) | [Message-ID](#) | [Return-Path](#) | [SPF](#) | [DKIM](#) | [DMARC](#) | [Authentication-Results](#) | [Email Spoofing](#)